STATISTICS – UNIT II

Descriptive Statistics | Data Science and Artificial Intelligence

Statistics – Unit 2: Descriptive Statistics

Course: Data Science and Artificial Intelligence

Unit: II – Descriptive Statistics

Topics: Central Tendency, Dispersion, Shape, and Outlier Detection

Purpose: Learning notes and Python implementation

1. Introduction to Descriptive Statistics

Descriptive statistics is the branch of statistics used to organise, summarise, analyse, and present the main features of a dataset. It helps us understand data before making comparisons or building Machine Learning models.

A good summary answers three questions:

1. **Centre:** What is a typical value?
2. **Spread:** How much do the values vary?
3. **Shape:** Is the distribution symmetric, skewed, or heavy-tailed? Are there unusual observations?

Descriptive statistics describes the data available; by itself, it does not prove causes or automatically generalise to a wider population.

Applications

- Summarising student marks and attendance.
- Understanding sales and business performance.
- Checking sensor readings and energy consumption.
- Exploring features before training an AI/ML model.
- Detecting unusual values that may be errors or important rare events.

2. Frequency Distributions and Histograms

A **frequency distribution** groups observations into values or class intervals and counts how often they occur.

- **Frequency (f):** Number of observations in a class.
- **Relative frequency:** Frequency divided by the total number of observations, f/n.
- **Cumulative frequency:** Running total of frequencies.
- **Histogram:** A chart using adjacent bars to show the frequency of numerical values grouped into bins.

Histograms help reveal the centre, spread, skewness, clusters, and possible outliers in a dataset. The appearance can change with bin width, so choose bins thoughtfully.

3. Measures of Central Tendency

Central tendency describes the centre or typical value of data.

3.1 Arithmetic Mean

The mean is the sum of all observations divided by the number of observations.

Sample mean: $\bar{x} = \Sigma x^i / n$

Population mean: $\mu = \Sigma x^i / N$

Example: For 10, 20, 30, mean = (10 + 20 + 30)/3 = 20.

Weighted mean: $\bar{x}_w = \Sigma(w^i x^i) / \Sigma w^i$

Use it when observations have different weights or importance.

Example: Internal score 70 with weight 0.4 and external score 85 with weight 0.6:

weighted mean = (70 × 0.4 + 85 × 0.6)/(0.4 + 0.6) = 79.

Properties: The mean uses every value, is useful for further calculations, and is sensitive to extreme values.

3.2 Median

The median is the middle observation after sorting the data.

- If n is odd, the median is the middle value.
- If n is even, the median is the average of the two middle values.

The median is resistant to extreme values and is often suitable for skewed data such as income or house prices.

Example: 4, 7, 9 → median = 7.

Example: 4, 7, 9, 12 → median = (7 + 9)/2 = 8.

3.3 Mode

The mode is the most frequently occurring value. A dataset can have one mode, more than one mode, or no repeated mode. Mode is especially useful for identifying the most common category.

Example: 2, 3, 3, 5, 8 → mode = 3.

3.4 Choosing a Measure

- **Mean:** Useful for reasonably symmetric numerical data without extreme outliers.
- **Median:** Useful when data is skewed or contains outliers.
- **Mode:** Useful for categories or the most common value.
- For a strongly skewed income dataset, median often represents a typical person better than mean.

4. Quartiles, Deciles, and Percentiles

These measures describe the position of observations in sorted data.

- **Q1 (first quartile):** Approximately the 25th percentile.
- **Q2 (second quartile):** Median, or 50th percentile.
- **Q3 (third quartile):** Approximately the 75th percentile.

- **Percentile:** A value below which a stated percentage of observations fall.
- **Deciles:** Divide sorted data into ten parts.

Different software packages can use different percentile interpolation methods, so state the method when exact quartile values matter.

5. Measures of Dispersion

Dispersion describes how spread out values are. Two datasets can have the same mean but very different variability.

5.1 Range

Range = Maximum – **Minimum**

Example: 10, 14, 18, 25 → range = 25 – 10 = 15.

Range is easy to calculate but depends only on the two extreme values.

5.2 Interquartile Range (IQR)

IQR = Q3 – **Q1**

IQR describes the spread of the middle 50% of observations and is more resistant to extreme values than range.

5.3 Five-Number Summary and Box Plot

The five-number summary consists of:

1. Minimum
2. Q1
3. Median (Q2)
4. Q3
5. Maximum

A box plot displays the median, quartiles, and spread. Values beyond the whiskers may be plotted individually as potential outliers, depending on the plotting convention.

5.4 Tukey Outlier Fences

Lower fence = Q1 – **1.5 × IQR**

Upper fence = Q3 + 1.5 × IQR

Values outside these fences are flagged as potential outliers. A flagged value is not automatically wrong; investigate its context.

5.5 Variance

Variance measures the average squared distance from the mean.

Population variance: $\sigma^2 = \Sigma(x^i - \mu)^2 / N$

Sample variance: $s^2 = \Sigma(x^i - x\blacksquare)^2 / (n - 1)$

The sample formula uses n – 1 (Bessel's correction) to estimate population variance without the usual downward bias under standard random-sampling assumptions.

5.6 Standard Deviation (SD)

Standard deviation is the square root of variance.

Population SD: $\sigma = \sqrt{\sigma^2}$

Sample SD: $s = \sqrt{s^2}$

SD is expressed in the original units of the data. It is sensitive to outliers because deviations are squared.

5.7 Mean Absolute Deviation (MAD)

Mean absolute deviation about the mean is the average absolute distance from the mean:

Mean absolute deviation = $\Sigma|x^i$ – x■| / n

Note: In outlier detection, MAD often means **median absolute deviation**, calculated as median($|x^i$ – median(x)|). This robust measure is different from mean absolute deviation.

6. The Empirical Rule

For data that follows an approximately normal, bell-shaped distribution:

- About **68%** of values lie within 1 standard deviation of the mean.
- About **95%** lie within 2 standard deviations.
- About **99.7%** lie within 3 standard deviations.

This rule is not appropriate for every distribution; check the shape first.

7. Relative Dispersion and Standardisation

7.1 Coefficient of Variation (CV)

CV = (Standard deviation / Mean) × 100%

CV expresses spread relative to the mean and can compare variability across datasets with different units, when the mean is meaningful and positive. It is not suitable in many cases where the mean is zero or near zero, or where values are on an arbitrary scale.

Example: Dataset A has mean 20 and SD 4 → CV = 20%. Dataset B has mean 50 and SD 5 → CV = 10%. Dataset A has greater relative variation.

7.2 Z-Score

A z-score describes how many standard deviations an observation is above or below the mean.

z = (x – μ**)/**σ for a population, or **z = (x** – **x■)/s** using sample estimates.

- z = 0: value is at the mean.
- z = +2: value is two SDs above the mean.

- z = –1: value is one SD below the mean.

A common rule flags |z| > 3 as a possible outlier, but this can be unreliable for skewed data or when outliers distort the mean and SD.

7.3 Feature Scaling in Machine Learning

- **Standardisation:** transforms a feature using its mean and standard deviation; values are often centred around 0 with SD around 1.
- **Min–Max scaling:** maps values to a selected range, commonly 0 to 1.
- **Robust scaling:** uses the median and IQR and is less affected by extreme values.

Fit scalers on training data only, then apply the fitted transformation to validation/test data to prevent data leakage.

8. Shape of a Distribution: Skewness

Skewness describes asymmetry in a distribution.

- **Approximately symmetric:** left and right sides are similar.
- **Positive (right) skew:** longer right tail; a few large values may pull the mean above the median.
- **Negative (left) skew:** longer left tail; a few small values may pull the mean below the median.

The relationship between mean and median is a useful clue, not a universal rule. A histogram or box plot should also be examined.

For strongly right-skewed positive data, possible approaches include reporting the median and IQR, investigating unusual values, or applying a suitable transformation such as log transformation. Choose based on the data and the purpose of analysis.

9. Kurtosis

Kurtosis describes tail heaviness and the tendency to produce extreme observations relative to a normal distribution. It is based on the fourth central moment.

Pearson kurtosis: $\beta_2 = \mu\blacksquare / \sigma\blacksquare$

Excess kurtosis: $\gamma_2 = \beta_2 - 3$

For a normal distribution, Pearson kurtosis is 3 and excess kurtosis is 0.

- **Positive excess kurtosis (leptokurtic):** heavier tails than the normal reference; extreme values may be more common.
- **Near-zero excess kurtosis (mesokurtic):** similar kurtosis to the normal reference.
- **Negative excess kurtosis (platykurtic):** lighter tails than the normal reference.

Software packages may use different conventions (Pearson kurtosis or excess kurtosis), so check the definition before interpreting a reported number.

10. Outlier Detection

An outlier is an observation unusually far from the rest of the data. It may be a data-entry error, sensor fault, rare event, fraud signal, or genuine special case.

Method 1: IQR / Tukey Fences

Flag $x < Q1 - 1.5(IQR)$ or $x > Q3 + 1.5(IQR)$. This method is robust and does not require a normal distribution.

Method 2: Z-Score

Flag values with $|z| > 3$ as a common screening rule. It works best as a rough check for data that is reasonably symmetric and can be distorted by outliers.

Method 3: Modified Z-Score

Calculate median absolute deviation:

$MAD = \text{median}(|x^i - \text{median}(x)|)$

Then:

$$\mathbf{M^i = 0.6745 \times (x^i - median(x)) / MAD}$$

A common rule flags $|M^i| > 3.5$. If MAD is zero, this formula needs special handling.

How to Handle an Outlier

1. Check whether it is a recording or measurement error.
2. Compare it with the original source and domain knowledge.
3. If it is valid, consider keeping it and choosing robust summaries/models.
4. If appropriate, correct, exclude, cap (winsorise), or transform the value with a documented reason.
5. Record the decision. Do not automatically delete outliers.

11. Sales Case Study

Monthly sales (in Rs. lakh):

18, 22, 19, 25, 24, 21, 23, 20, 26, 22, 60, 22

The value 60 represents a possible bulk order and is a potential outlier.

Sorted data:

18, 19, 20, 21, 22, 22, 22, 23, 24, 25, 26, 60

Using the common median-of-halves quartile method:

- n = 12
- Median = (22 + 22) / 2 = 22
- Q1 = median of lower half (18, 19, 20, 21, 22, 22) = (20 + 21)/2 = 20.5
- Q3 = median of upper half (22, 23, 24, 25, 26, 60) = (24 + 25)/2 = 24.5
- IQR = 24.5 – 20.5 = 4
- Lower fence = 20.5 – 1.5(4) = 14.5
- Upper fence = 24.5 + 1.5(4) = 30.5

Therefore, 60 is flagged as a potential outlier using this quartile convention. Quartile values may differ slightly with other software conventions.

The mean is pulled upward by 60, while the median is much less affected. This illustrates why both centre and spread should be examined.

12. Python for Descriptive Statistics

The accompanying Python script uses NumPy, Pandas, SciPy, and Matplotlib to:

- Calculate mean, median, mode, and weighted mean.
- Calculate range, quartiles, IQR, variance, SD, MAD, CV, and z-scores.
- Calculate skewness and kurtosis.
- Flag potential outliers using IQR and modified z-score methods.
- Create a histogram and box plot.
- Analyse the monthly sales case study.

Install dependencies

```
python -m pip install -r requirements.txt
```

Run the script

```
python descriptive_statistics_unit2.py
```

13. Conclusion

Descriptive statistics turns raw data into meaningful information by summarising centre, spread, and shape. Mean, median, and mode describe central tendency; range, IQR, variance, and standard deviation describe dispersion; skewness and kurtosis describe distribution shape. Outlier detection helps identify unusual observations for investigation.

These techniques are essential in exploratory data analysis (EDA), data cleaning, feature preprocessing, and Machine Learning. Statistical results should always be interpreted in context, and potential outliers should not be removed without justification.

Learning Outcomes

After studying Unit II, a learner should be able to:

1. Explain the purpose of descriptive statistics.
2. Construct and interpret frequency distributions and histograms.
3. Calculate mean, weighted mean, median, and mode.
4. Calculate range, quartiles, IQR, variance, and standard deviation.
5. Interpret the five-number summary and box plots.
6. Explain the empirical rule and coefficient of variation.
7. Calculate and interpret z-scores.
8. Explain skewness and kurtosis.
9. Detect and investigate potential outliers.
10. Apply descriptive statistics using Python.

Source

Learning summary prepared with reference to the Unit II course material, *Descriptive Statistics for Data Science & Artificial Intelligence*, by Dr. S. Satyanarayana, Sanjivani University.

Note: These notes summarise key topics for learning and are not a replacement for the complete course material.